

HUMBER ENGINEERING

MENG 3020

SYSTEMS MODELING & SIMULATION

LECTURE 1

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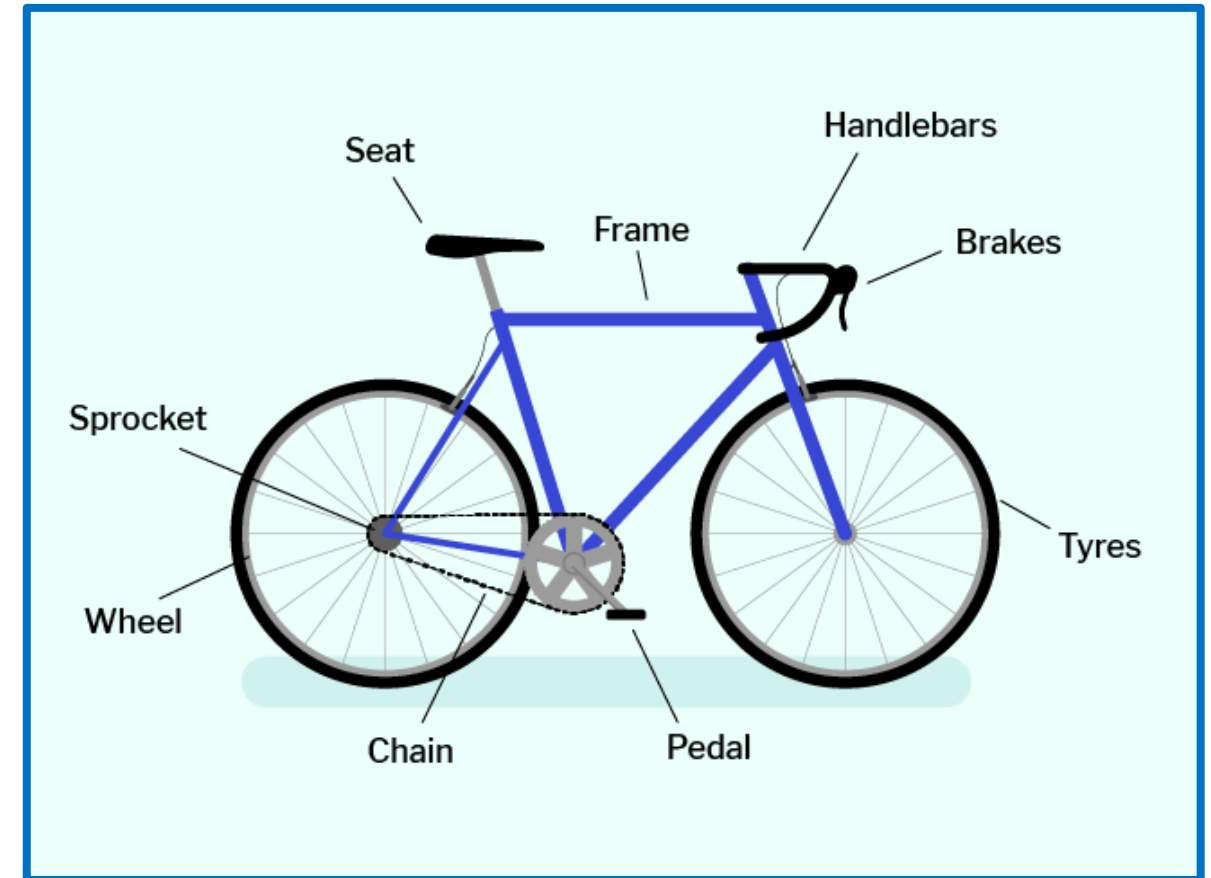
Introduction to System Modeling

- Basic Definitions & Terminologies
- Static vs. Dynamic Systems
- Linear Time-Invariant Systems
- Linearization
- Modeling of Dynamic Systems
- System Modeling Procedure
- Simulation of Dynamic Systems

Basic Definitions & Terminologies

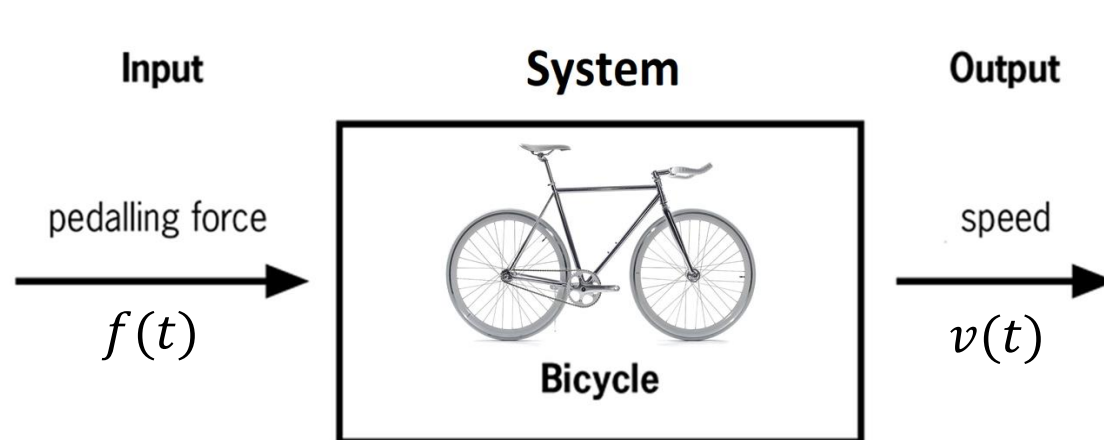
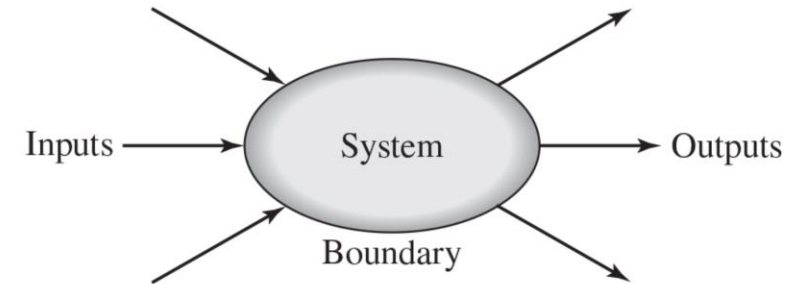
□ What is a System?

- Engineers are interested in analyzing and designing **physical systems**.
- A **system** is a combination of components or subsystems acting together to perform a specific objective.
 - The **component** is a single function unit of a system.
- **Examples of systems:**
 - Home Heating System
 - Industrial Robot
 - Computer
 - Mobile Phone
 - Human Body
 - Aircraft / Spacecraft
 - Automobile / Bicycle
 - Washing Machine
 - Road Traffic System
 - Stock Market
 -



Basic Definitions & Terminologies

- Systems have inputs and outputs
 - System characteristics determine how inputs translate to outputs.
- Inputs:** Any causes or excitations acting on the system from the world external to drive the system.
 - Applied force or torque in mechanical systems
 - Applied voltage or current in electrical systems
 - Applied pressure in hydraulic or pneumatic systems
- Outputs:** Variables of interest to be observed or measured by sensors to assess the dynamic condition of the system.
 - Speed or position of a car
 - Voltage or current of a resistor
 - Rate at which a liquid flows through a pipe
- State Variables:** Variables to represent the internal status or memory of the system, that are used to mathematically model the dynamic behavior of the system.



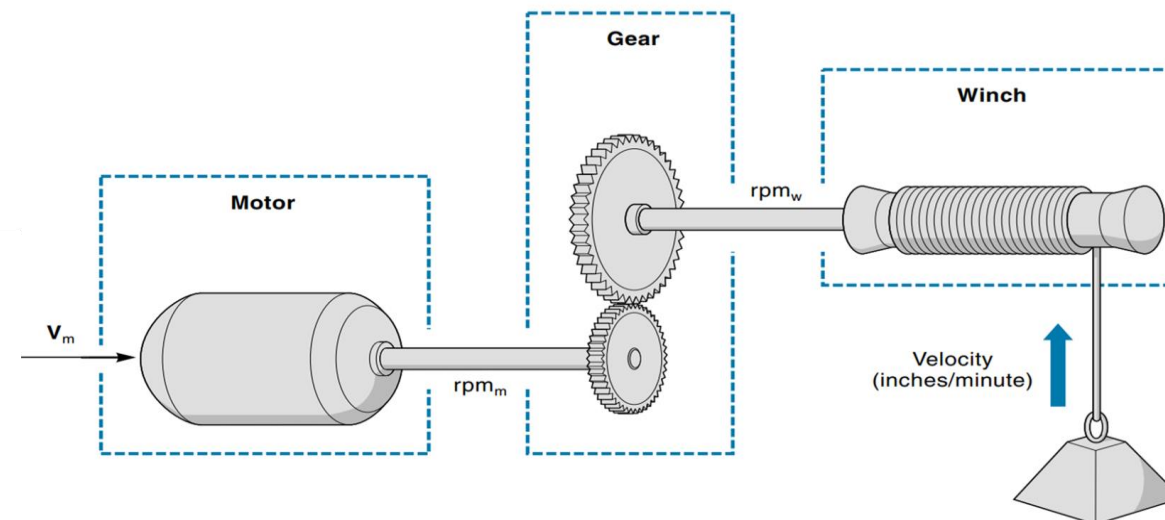
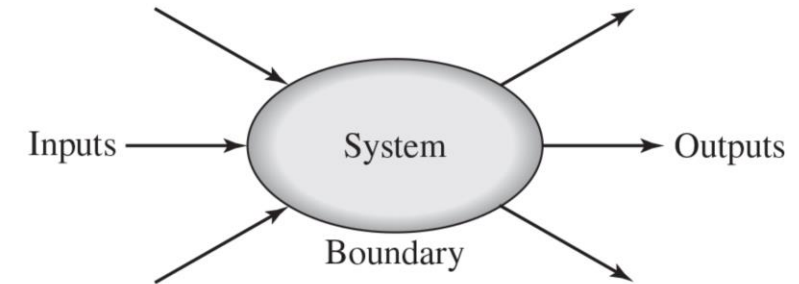
State Variables: Handlebars angle, Acceleration



State Variables: DC Motor speed, Arm rotation angle

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- **State Variables:** Variables to represent the internal status or memory of the system, that are used to mathematically model the dynamic behavior of the system.
- There is a **cause-and-effect** relationship between the system components and the systems and the world external.



Basic Definitions & Terminologies

□ What is a System Model?

- In order to **study** and **analyze** the systems we have to be able to describe these systems in a tractable, **mathematical** way.
- We represent these systems with **models**.
- **System Model** is a **description of the system**, which captures its essential behavior and represent its main dynamics.
 - It is an abstract representation of the system
 - Captures some aspects of the real system's behavior – the behavior we care about – while ignoring others
 - Simplified in some way: Smaller, Less complex, Linear, Lossless, etc. ...
- Model of a physical system may be:
 - A **simplified physical system** itself
 - For example, a scaled model for wind-tunnel testing, a road vibration test machine
 - A **mathematical model**
 - An **equation** or **system of equations** that describe the aspects of system behavior that interest us (while ignoring others)



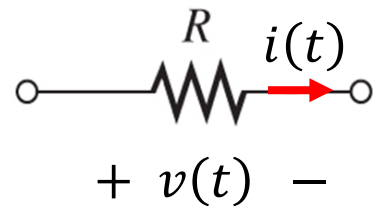
Basic Definitions & Terminologies

□ Why do we need the system model?

- We use **models** for *simulation* and *analysis* of the system and *design* a control system.
- *Simulate* to predict the dynamic response of the components and overall system to different types of inputs
 - Could be *physical simulation* or *mathematical simulation*
 - We are interested in mathematical simulation, which could be either **analytical** or **numerical**
- *Analysis* of system behavior to understand how components affect each other and the overall system,
 - Helpful in fault detection and maintenance
- *Design* a control system to improve and enhance the dynamic response as desired
 - Modifying the system parameters to achieve desired behavior
 - Control system design – adding feedback and a controller to the system to improve system performance
 - This is the subject of the **MENG 3510 – Control Systems** course for next semester.
- In this course we focus on developing Linear Time-Invariant (LTI) models of dynamic systems.

Static System / Dynamic System

- **Static System:** Present value of output depends only on the present value of the input.
 - *Static systems are also called **instantaneous systems**.*
 - *They have **no memory (no energy storage element)***
 - *These systems are modeled by **algebraic equations**.*
- For example, a **resistor** is a **static element**. Assuming the current as the input and the voltage as the output.
- The current flowing through a resistor depends only on the present value of the applied voltage.
- If the current changes, the voltage will also change instantaneously.



$$v(t) = Ri(t)$$

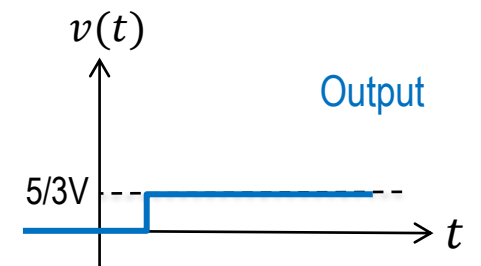
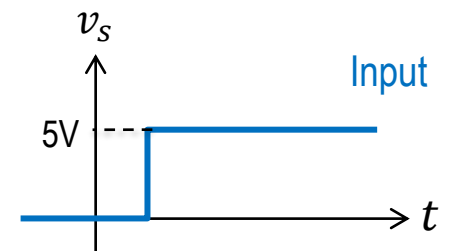
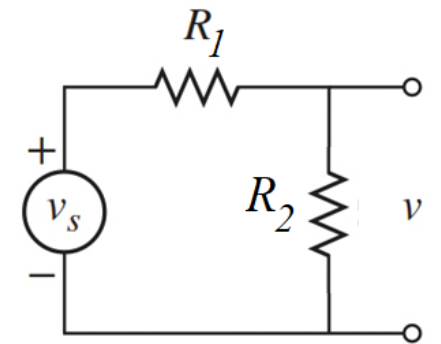
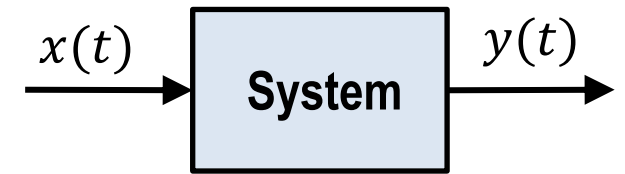
- A **purely resistive circuit** is a **static system**.

Assume that the supplied voltage v_s is the **input** and voltage across resistor R_2 is the **output**.

The input-output relationship is:

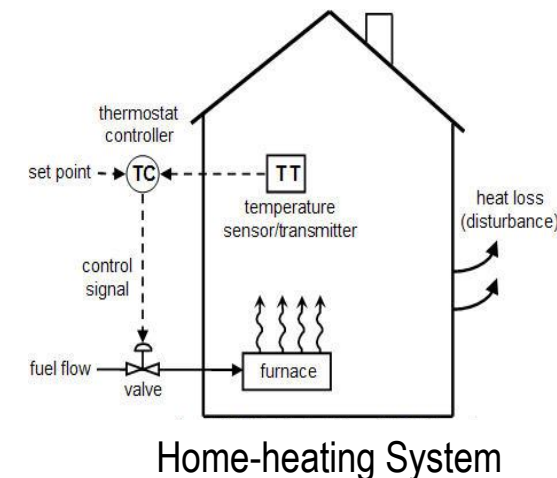
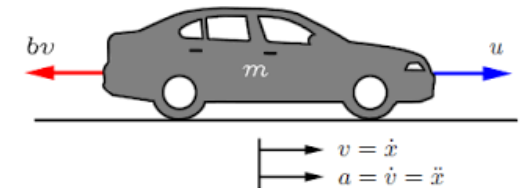
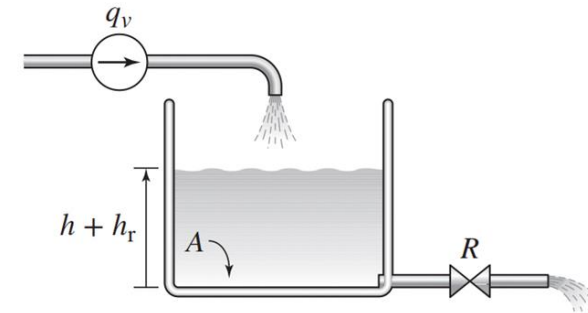
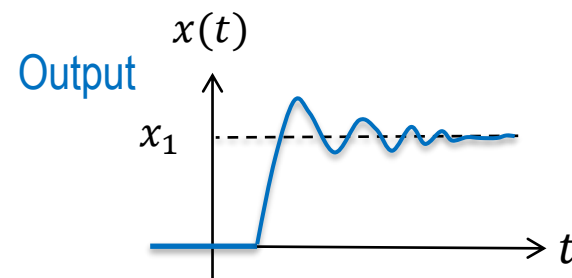
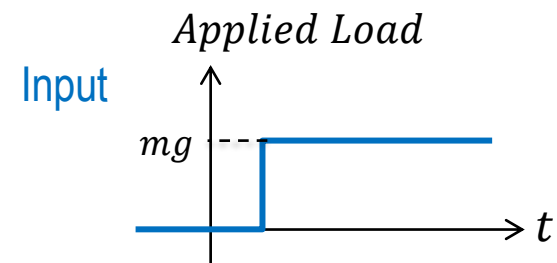
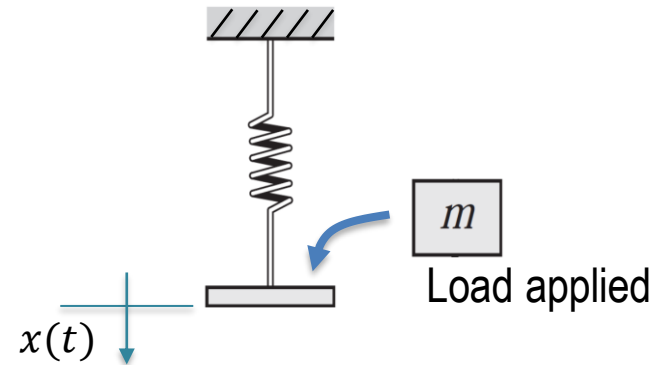
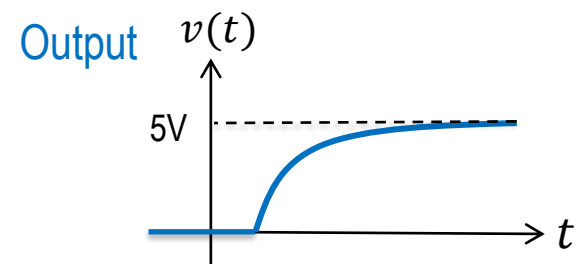
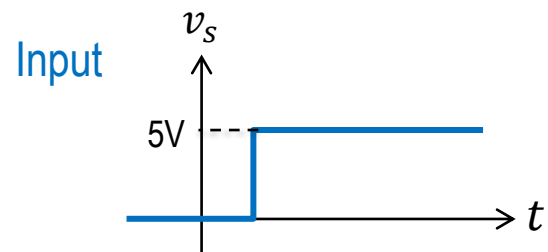
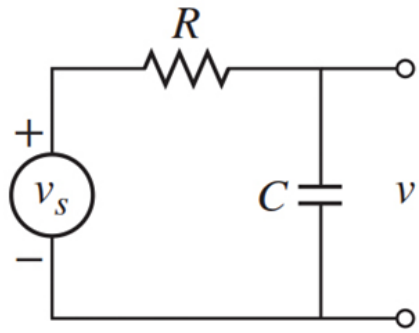
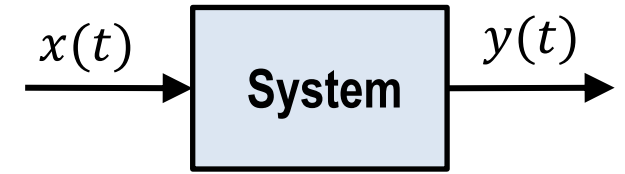
$$v(t) = \frac{R_2}{R_1 + R_2} v_s(t)$$

Assume $R_1 = 100k\Omega$, $R_2 = 50k\Omega \rightarrow v(t) = \frac{1}{3} v_s(t)$

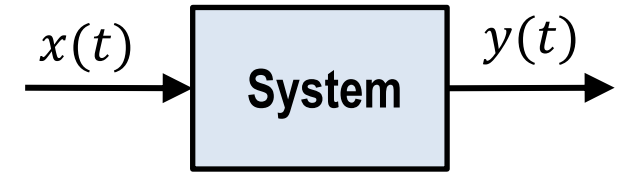


Static System / Dynamic System

- **Dynamic System:** Present value of output depends on the **past** and **present** value of the input.
 - **Dynamic systems have *memory (energy storage element)***
 - **These systems are modeled by *differential equations*.**
- For example, RLC circuits, an aircraft, an automobile, a robot, home heating system, mass-spring balanced system, etc. ...



Linear Time-Invariant System



- **Time-Invariant System:** A system whose **parameters** are **constant** and do not change with time.
 - If the system parameters are function of time, then the system is called a **time-varying system**.
 - For example, a **rocket** or an **airplane** is a **time-varying system**, where its mass changes dynamically as it burns fuel, altering its dynamics over time.
- **Linear Time-Invariant (LTI) System:** A system whose output is **proportional** to its input.
 - **LTI dynamic** systems can be described by **Ordinary Differential Equations (ODE)** or **linear differential equations**.
 - For example,
$$a\ddot{y}(t) + b\dot{y}(t) + cy(t) = x(t)$$
 - **Nonlinear** dynamic systems are described by **Nonlinear Differential Equations**.
 - For example,
$$a\ddot{y}(t) + b \cdot \ln(\dot{y}(t)) + cy^2(t) = x(t)$$
 - Truly linear systems do not exist in reality; all systems are inherently **nonlinear**.
 - However, there are some reasons why we are interested of **linear models** of dynamic systems.

❖ Differential equations based on linear models are easier to solve than the nonlinear models, which simplifies modeling and analysis.

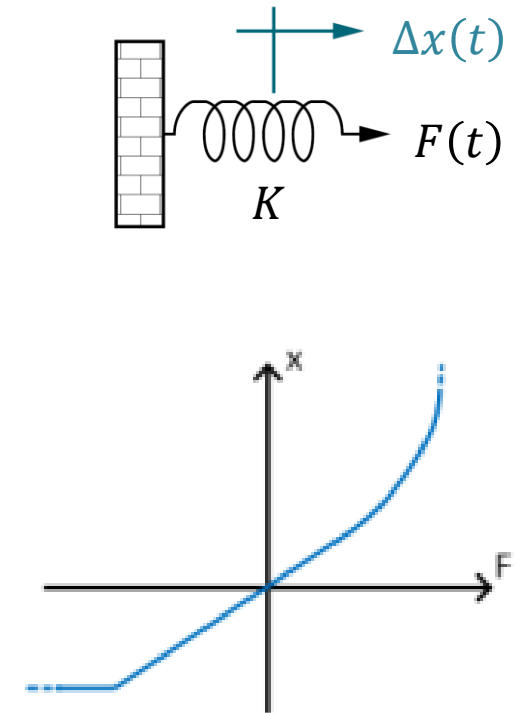
- Therefore, when developing models, we try to obtain a **linear model** whenever possible, but sometimes it may result to loss of accuracy.

Linear Time-Invariant System

❖ Many systems can be accurately modeled as linear over a small enough range.

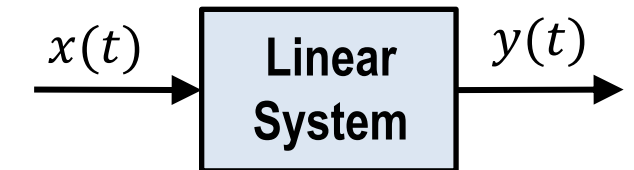
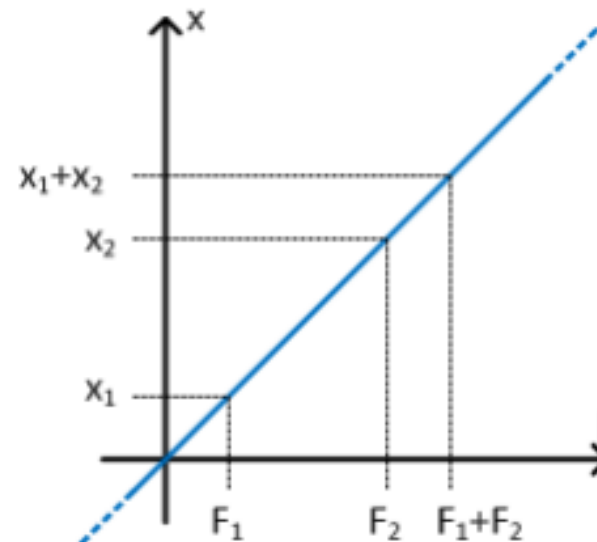
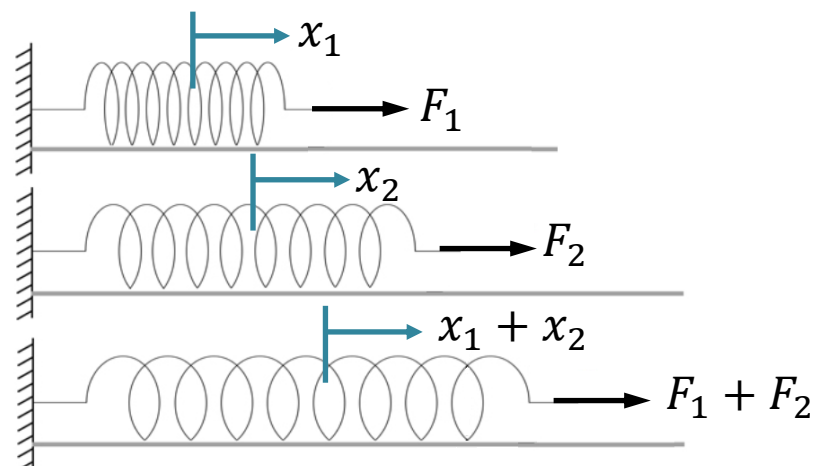
- For example, consider a simple spring.
 - Assuming the applied force $f(t)$ as an input and the displacement $\Delta x(t)$ of the spring as the output.
 - Is the spring a linear component?
 - No** – Over a full range of force and displacement, it is clearly nonlinear.
 - Yes** – For small values of force and displacement the spring is accurately approximated as linear
 - Obeys the Hooke's law,

$$F(t) = K\Delta x(t)$$



❖ The superposition and homogeneity principles apply to linear systems.

- For example, consider a linear spring.



$$\begin{array}{lcl} x_1 & \rightarrow & y_1 \\ x_2 & \rightarrow & y_2 \\ \alpha x_1 + \beta x_2 & \rightarrow & \alpha y_1 + \beta y_2 \end{array}$$

Superposition & Homogeneity Principles

Linearization

- A **nonlinear system** can be approximated by a **linear model** for small deviations around an **operating point** of the system using the **Taylor Series** expansion by neglecting the higher-order terms.
- First, we will see how to linearize a **function**; later, we will apply the method to the linearization of a **differential equation**.

□ Linearization of a Nonlinear Function

- Assume a nonlinear system with input x and output $f(x)$ operating at point A
- If the slope of the curve at point A is m_A .
- Small changes in the input can be related to changes in the output about the point A by way of the slope of the curve, m_A , at point A .

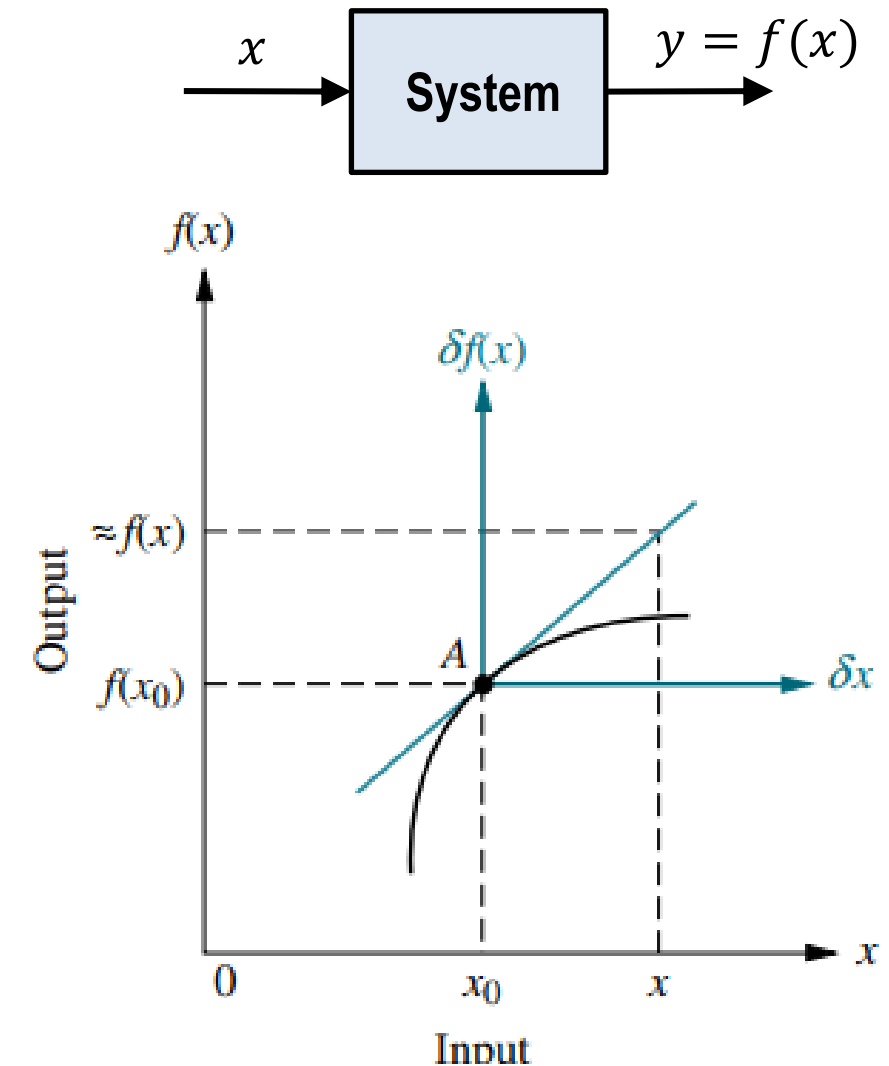
$$|f(x) - f(x_0)| \approx m_A(x - x_0)$$

$$\delta f(x) \approx m_A \delta x$$

$$f(x) \approx f(x_0) + m_A(x - x_0)$$

where the slope m_A is determined as:

$$m_A = \left. \frac{df(x)}{dx} \right|_{x=x_0}$$



Linearization

□ Linearization – Example 1

- Linearize a system represented by $f(x) = 5 \cos x$ about $x = \pi/2$.
- First find the **derivative** of $f(x)$

$$\frac{df(x)}{dx} = -5 \sin x$$

- Evaluate the derivative at point $x = \pi/2$ to find the **slope**.

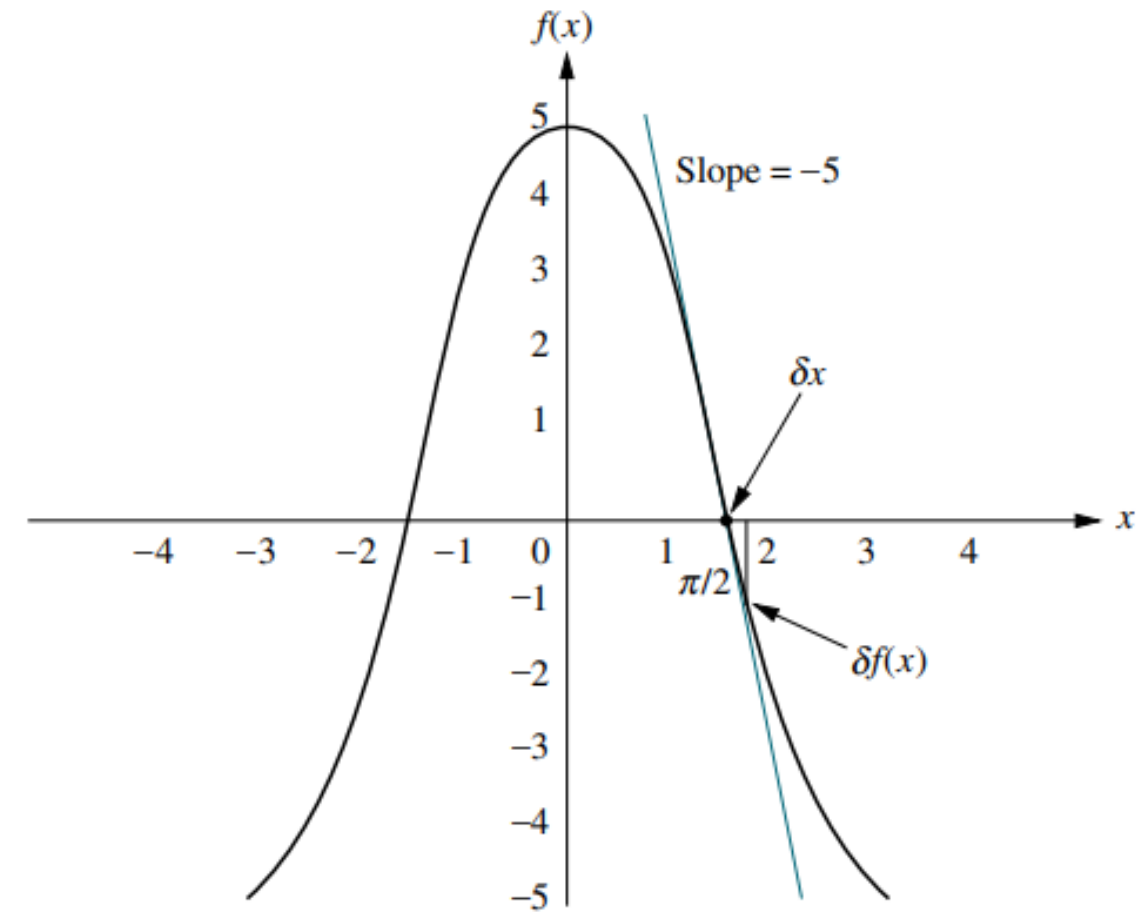
$$m_A = \left. \frac{df(x)}{dx} \right|_{x=\frac{\pi}{2}} = -5 \sin \frac{\pi}{2} = -5$$

- Evaluate the $f(x)$ at $x = \pi/2$ to find the **slope**.

$$f(x_0) = f\left(\frac{\pi}{2}\right) = 5 \cos \frac{\pi}{2} = 0$$

- Thus, the linear approximation of the system around point $x = \pi/2$ is:

$$\delta f(x) = -5\delta x \quad \rightarrow \quad f(x) = -5\left(x - \frac{\pi}{2}\right)$$



$$f(x) \approx f(x_0) + m_A(x - x_0)$$

$$m_A = \left. \frac{df(x)}{dx} \right|_{x=x_0}$$

Linearization

□ Linearization – Example 2

- Linearize a dynamic system represented by nonlinear equation for small excursions about $x = \frac{\pi}{4}$.
- Small excursion about $x = \frac{\pi}{4}$ is $\rightarrow \delta x = x - \frac{\pi}{4}$
- Set $x = \delta x + \frac{\pi}{4}$ into the equation

$$\frac{d^2 \left(\delta x + \frac{\pi}{4} \right)}{dt} + 2 \frac{d \left(\delta x + \frac{\pi}{4} \right)}{dt} + \cos \left(\delta x + \frac{\pi}{4} \right) = 0 \rightarrow \frac{d^2 \delta x}{dt} + 2 \frac{d \delta x}{dt} + \cos \left(\delta x + \frac{\pi}{4} \right) = 0$$

- Linearized the term $\cos \left(\delta x + \frac{\pi}{4} \right)$ around $x = \frac{\pi}{4}$ with the [Taylor series approximation](#).

$$f(x) = \cos \left(\delta x + \frac{\pi}{4} \right) \rightarrow f(x_0) = f \left(\frac{\pi}{4} \right) = \cos \frac{\pi}{4} = \frac{\sqrt{2}}{2} \Rightarrow f(x) = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} \left(x - \frac{\pi}{4} \right) = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} \delta x$$
$$\frac{df(x)}{dx} = -\sin x \rightarrow m_A = -\sin \frac{\pi}{4} = -\frac{\sqrt{2}}{2}$$

- Thus, the linear approximation of the differential equation around point $x = \pi/4$ is:

Solve for δx and find $x = \delta x + \pi/4$

$$\frac{d^2 \delta x}{dt} + 2 \frac{d \delta x}{dt} - \frac{\sqrt{2}}{2} \delta x = -\frac{\sqrt{2}}{2}$$

$$\frac{d^2 x}{dt} + 2 \frac{dx}{dt} + \cos x = 0$$

$$f(x) \approx f(x_0) + m_A(x - x_0)$$

$$m_A = \left. \frac{df(x)}{dx} \right|_{x=x_0}$$

Modeling of Dynamic Systems

□ Lump-Parametric Modeling

- Find the input-output relationships for systems by considering them to be composed of just a few **simple basic elements** and applying the **physical laws** from **first-principles**.
- **Electrical systems** can be considered as compose of basic elements, which can be represented by **resistors**, **capacitors**, **inductors** and **op-amps**.
 - These are characterized by voltage –current relationships for components and the laws of interconnection **Kirchhoff's Voltage Law (KVL)** and **Current Law (KCL)**
- **Mechanical systems** can be considered as compose of basic elements, which can be represented by **springs**, **dampers** and **masses**, and are characterized by **Newton's Laws of motion**.
- Mainly used to model **electrical**, **mechanical**, and **electromechanical** systems.
- The model is obtained as a **differential equation**, then shown in any standard form of, **Transfer Function model**, **State-Space model** or **Block diagram model**.

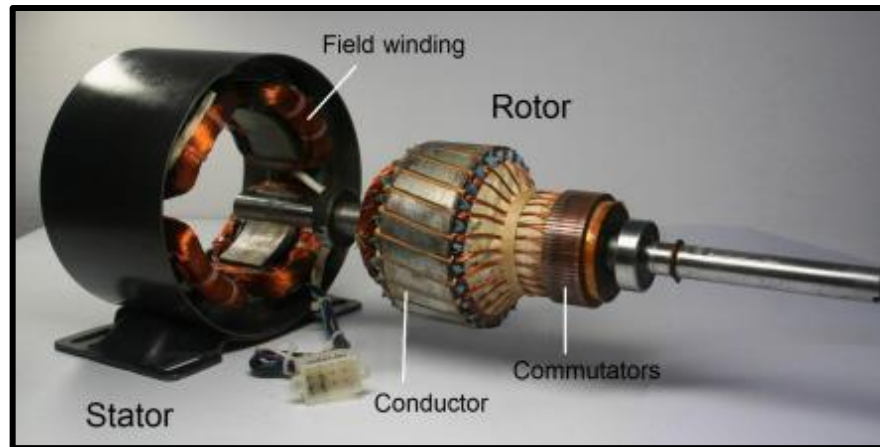
□ Empirical Modeling & System Identification

- An **experimental approach**, which mainly used to model **thermal** and **fluid (hydraulic and pneumatic)** processes.
- Some **experiments** are performed on the system to **collect input-output data**, a model is then fitted to the collected data by assigning suitable numerical values to its parameters.

Modeling of Dynamic Systems

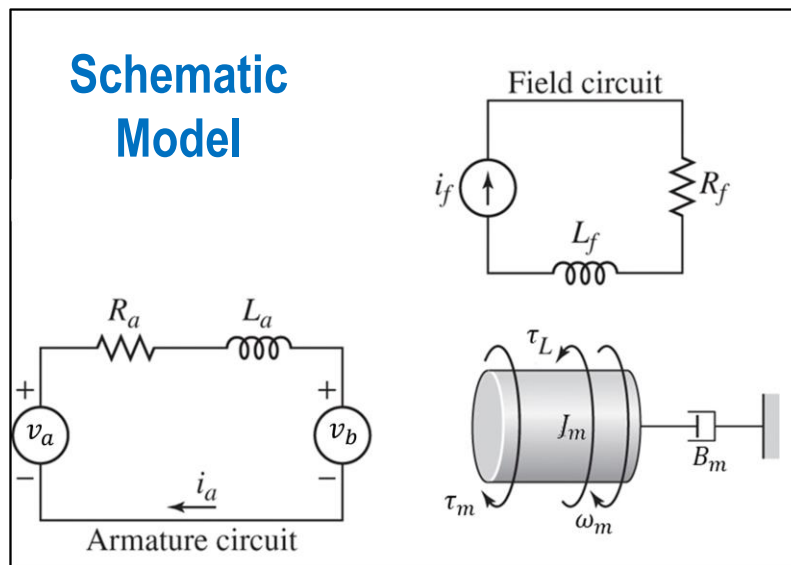
- The **mathematical models** are obtained as differential equations by applying the **physical laws** from **first-principles**.

○ Armature Controlled DC Motor



Input: Applied armature voltage

Output: Angular speed of rotor shaft



- Divide the system into idealized components
- Apply physical laws to the elements
- Apply interconnection laws between elements
- Combine the equations to obtain a standard form models
 - Transfer Function Model
 - State-Space Model
 - Block Diagram Model

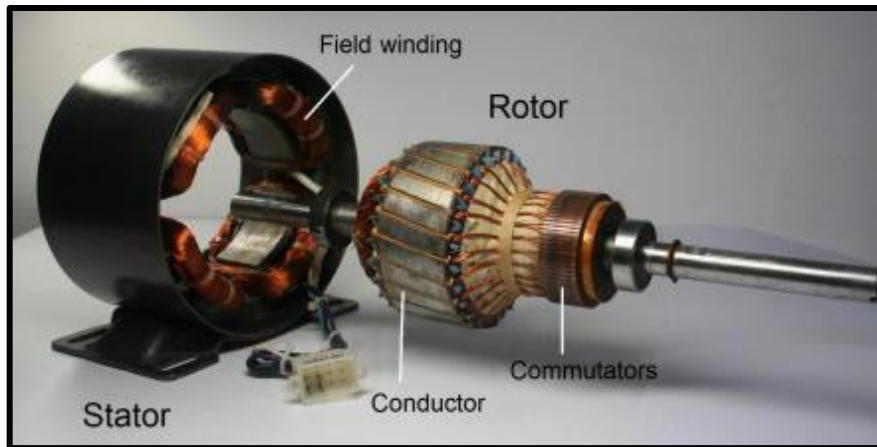
Differential Equations

$$\begin{cases} v_a(t) = R_a i_a(t) + L_a \frac{di_a(t)}{dt} + v_b(t) \\ \tau_m(t) = K_i i_a(t) \\ \tau_m(t) = J_m \frac{d\omega_m(t)}{dt} + B_m \omega_m(t) + \tau_L(t) \\ v_b(t) = K_b \omega_m(t) \end{cases}$$

Modeling of Dynamic Systems

- The **mathematical models** are obtained by performing some **experiments** and collecting **input-output data**.

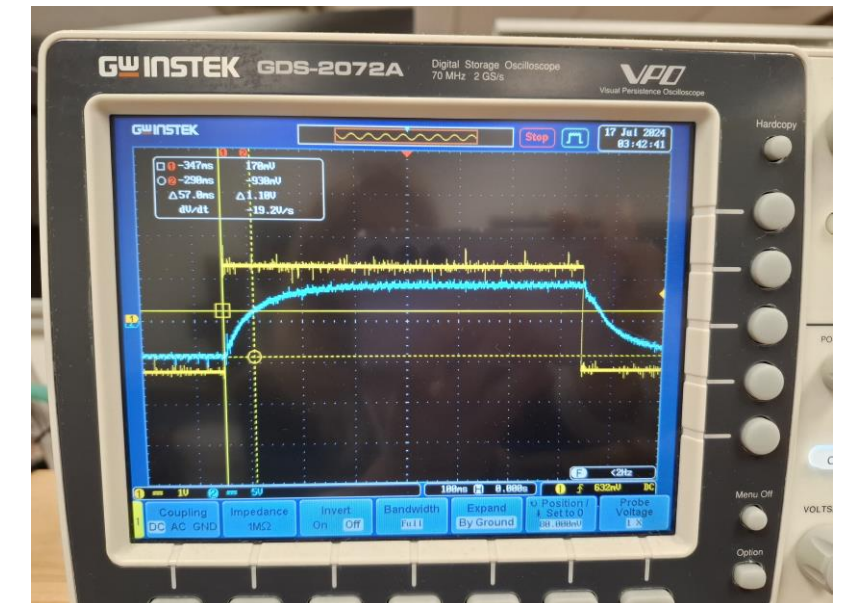
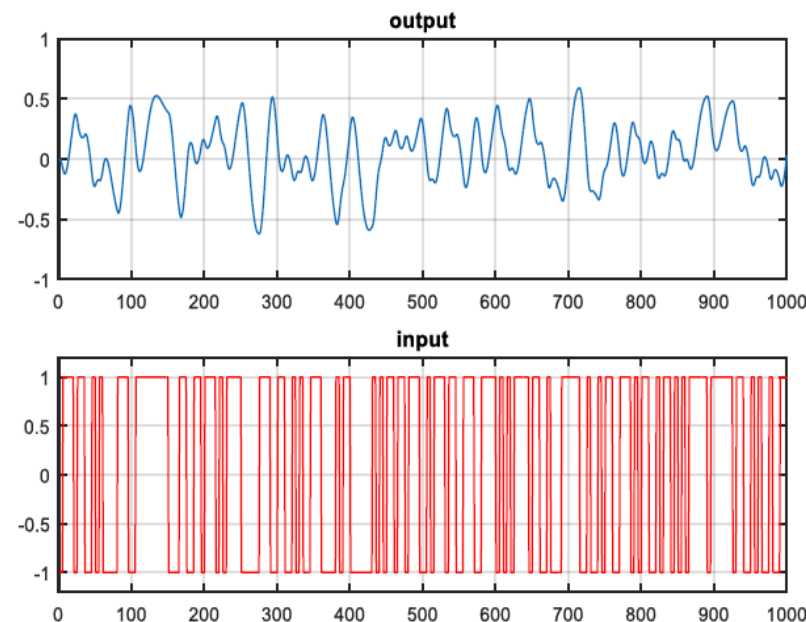
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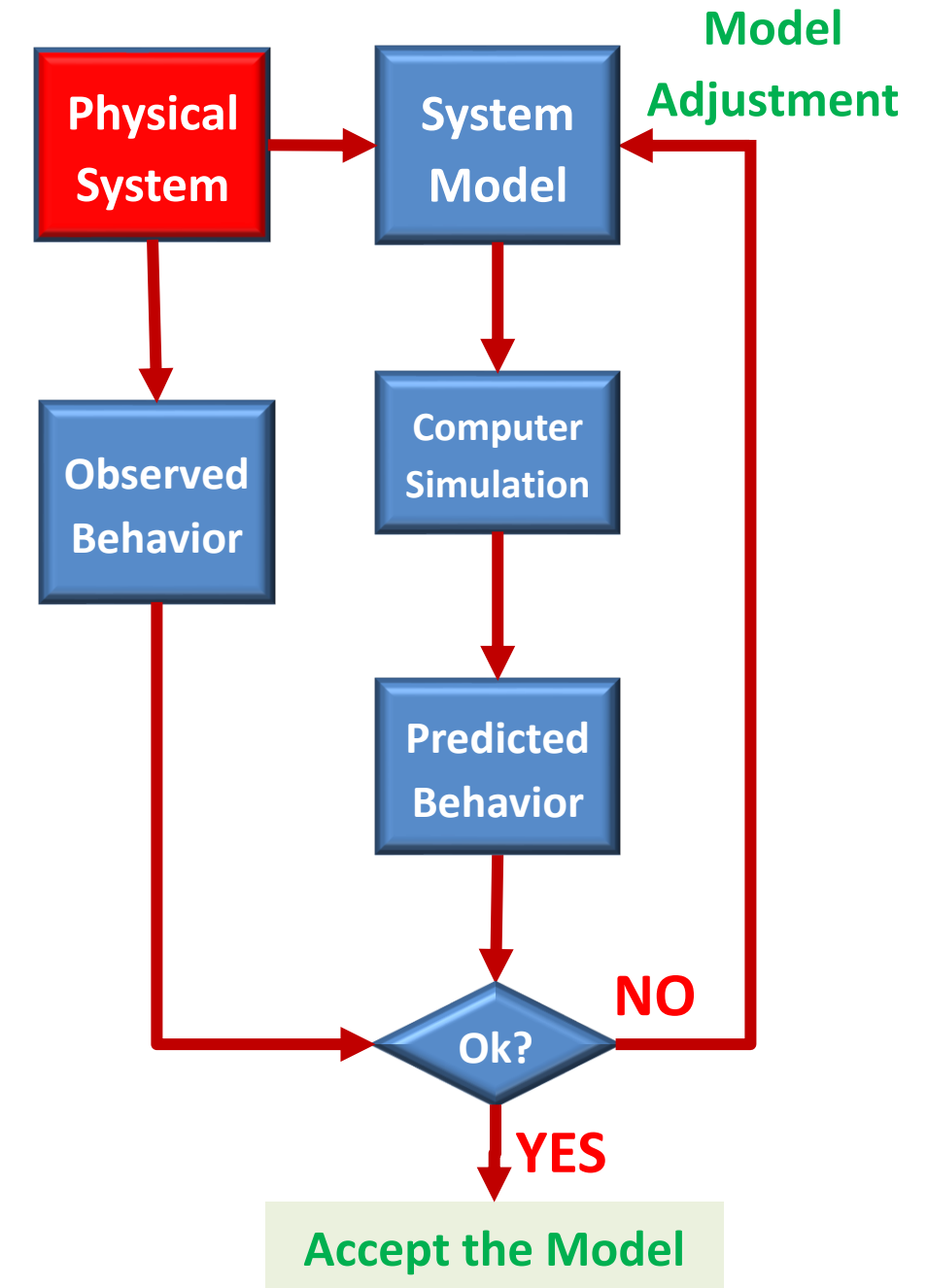
- Apply the appropriate input data.
- Collect the output data
- Match the output data with a standard model
 - Transfer Function Model**
- Estimate the model parameters from the input-output data



- Data-driven **System Identification** approach.
- Based on **statistical** methods.

System Modeling Procedure

- ✓ **Define the purpose or objective of the model**
 - Identify system boundaries, subsystems & components, interconnecting variables, inputs and outputs.
- ✓ **Determine the model for each component or subsystem**
 - Apply the known physical laws for obtaining differential equations, transfer function or state-variable equations, when possible, otherwise use experimental data to identify input-output relationships.
- ✓ **Integrate the subsystem models into an overall system model**
 - Combine equations, eliminate common variables, check for sufficient equations to solve the system.
- ✓ **Verify the model validity and accuracy**
 - Implement a simulation of the model equations and compared with experimental data for the same conditions.
- ✓ **Make simplifications to create an approximate model suitable for design**
 - Linearization of model equations
 - Reduce the order of the model by eliminating unimportant dynamics

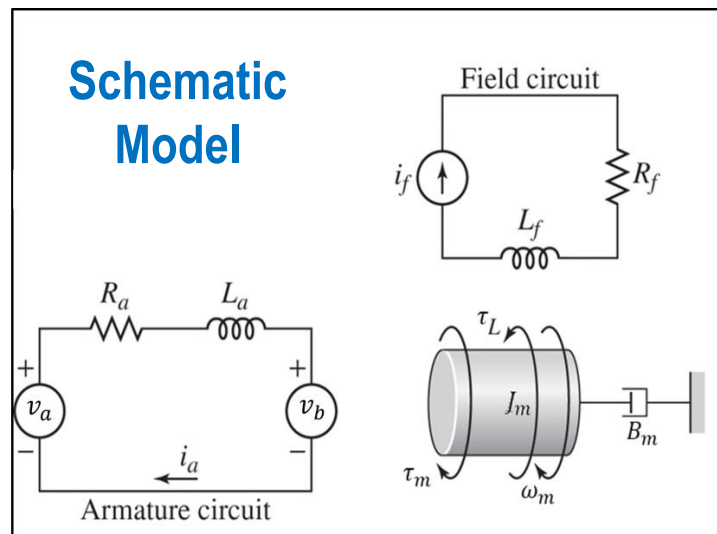
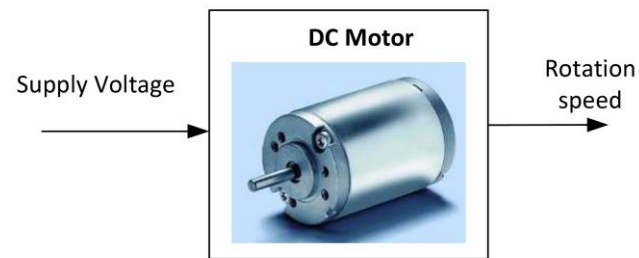


Simulation of Dynamic Systems

- The **simulation technique** includes an extensive collection of methods and applications aimed at **reproducing the actual behavior of nonlinear systems** on a digital computer with appropriate simulation software.
- We will particularize the computer simulation task by employing simulation software such as **MATLAB**, **Simulink** or **Simscape**, using codes, block diagrams and physical elements interconnected to synthesize the model equations.
- A **computer simulation** must embody several components:
 - In first place, the **structure of the dynamic model** to be simulated must be known, as also the set of model parameters and initial conditions.
 - In second place, the set of **input signals** must also be embodied, and a set of **output signals** must be explicitly defined in order to follow the system evolution.
 - Finally, a **simulation run time** must be included in order to select the numerical integration method used and the value of its associated parameters, integration step, error tolerances, etc.

System Simulation Tools: MATLAB

- MATLAB** is a high-level technical computing environment suitable for solving scientific and engineering problems.



$$\left\{ \begin{array}{l} v_a(t) = R_a i_a(t) + L_a \frac{di_a(t)}{dt} + v_b(t) \longrightarrow V_a(s) = (L_a s + R_a) I_a(s) + V_b(s) \\ \tau_m(t) = K_i i_a(t) \longrightarrow T_m(s) = K_i I_a(s) \\ \tau_m(t) = J_m \frac{d\omega_m(t)}{dt} + B_m \omega_m(t) + \tau_L(t) \longrightarrow T_m(s) = (J_m s + B_m) \Omega(s) + T_L(s) \\ v_b(t) = K_b \omega_m(t) \longrightarrow V(s) = K_b \Omega(s) \end{array} \right.$$

Differential Equations



Transfer
Function Model

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_i}{L_a J_m s^2 + (L_a B_m + J_m R_a) s + R_a B_m + K_i K_b}$$



MATLAB
Code

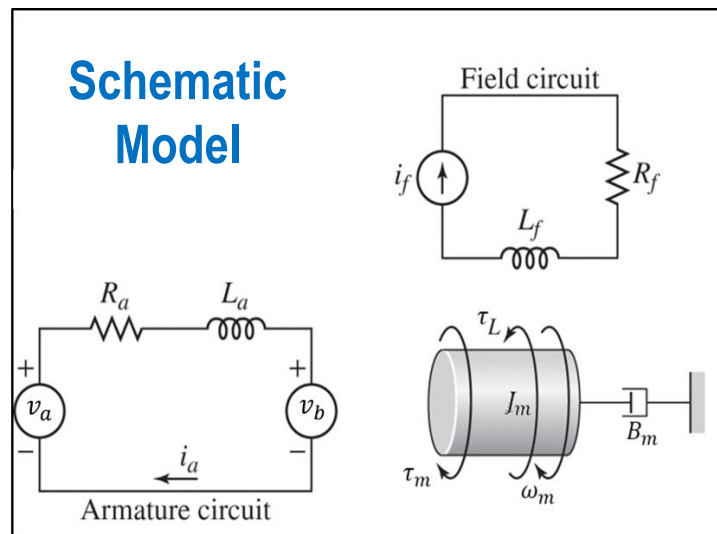
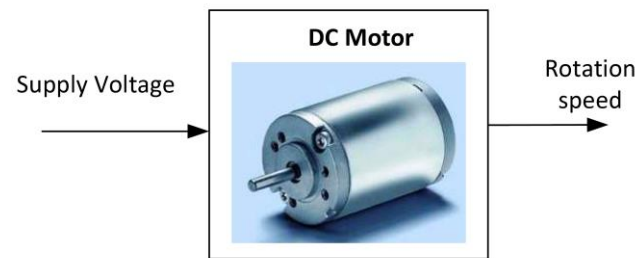
```
Ki = 0.05; Kb = Ki;
La = 2e-3; Ra = 0.5;
Jm = 9e-5; Bm = 1e-4;

speed = tf(Ki, [La*Jm, La*Bm+Jm*Ra, Ra*Bm+Ki*Kb]);

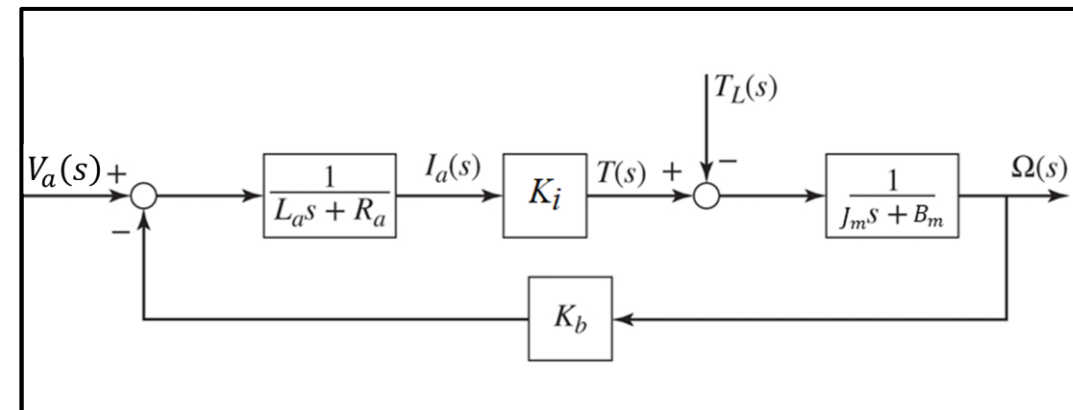
step(speed);
```


System Simulation Tools: Simulink

- Simulink** is an extension to MATLAB that allows users to rapidly and accurately build computer models of dynamical systems using **block diagrams**.



$$\left\{ \begin{array}{l} v_a(t) = R_a i_a(t) + L_a \frac{di_a(t)}{dt} + v_b(t) \longrightarrow V_a(s) = (L_a s + R_a) I_a(s) + V_b(s) \\ \tau_m(t) = K_i i_a(t) \longrightarrow T_m(s) = K_i I_a(s) \\ \tau_m(t) = J_m \frac{d\omega_m(t)}{dt} + B_m \omega_m(t) + \tau_L(t) \longrightarrow T_m(s) = (J_m s + B_m) \Omega(s) + T_L(s) \\ v_b(t) = K_b \omega_m(t) \longrightarrow V_b(s) = K_b \Omega(s) \end{array} \right.$$



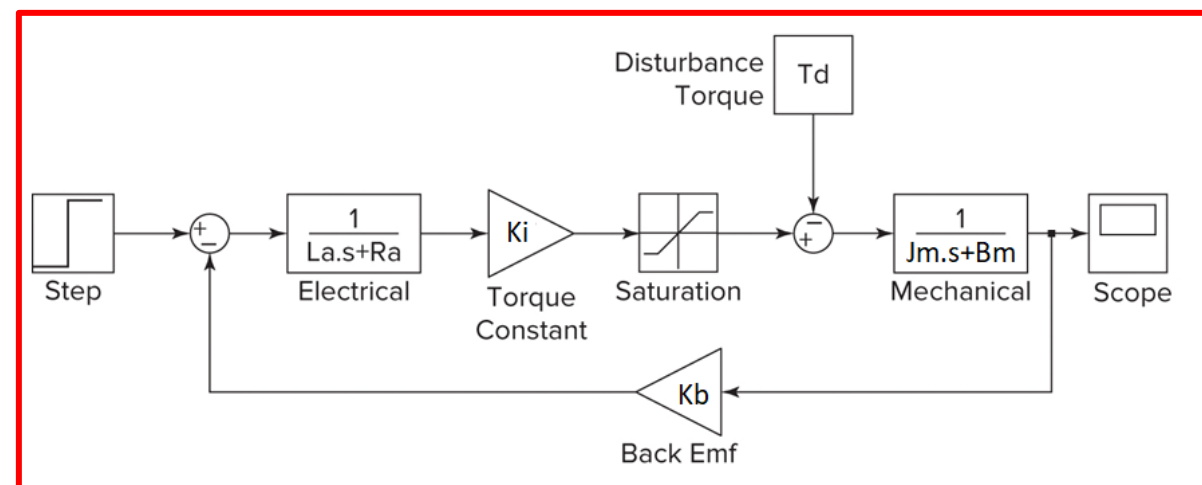
Differential Equations



Block Diagram Model

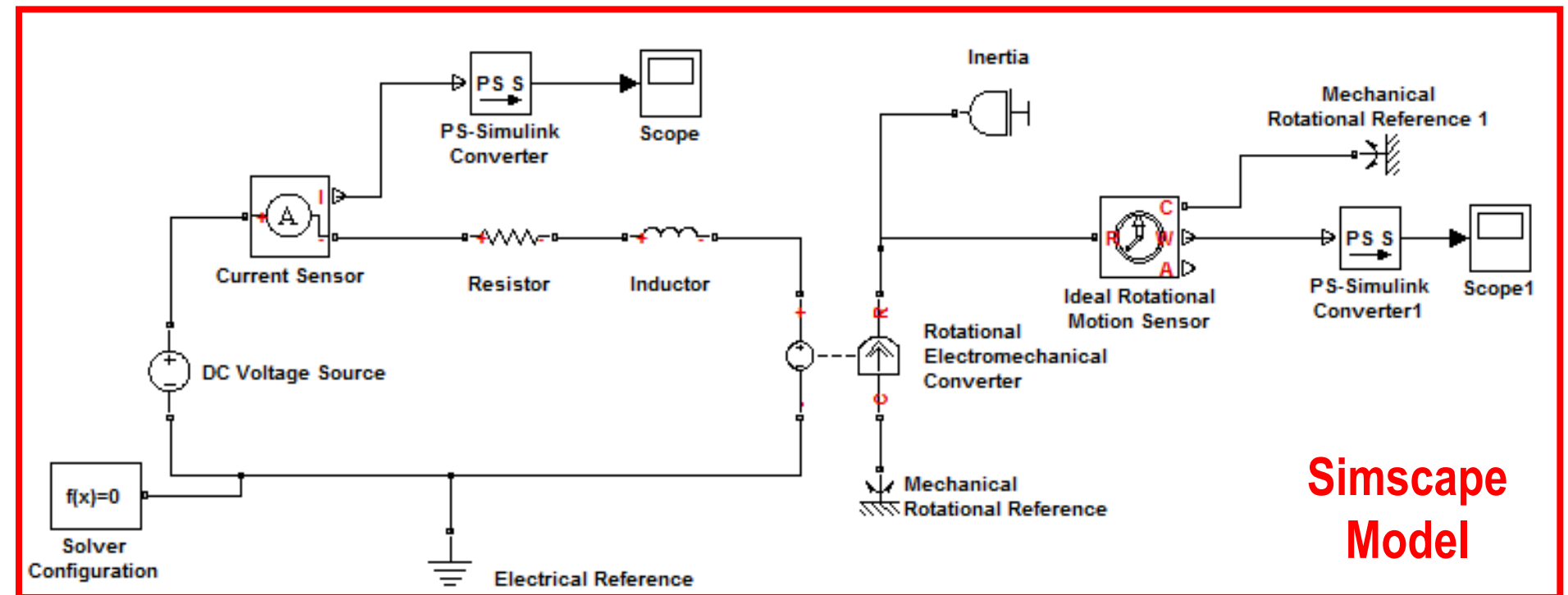
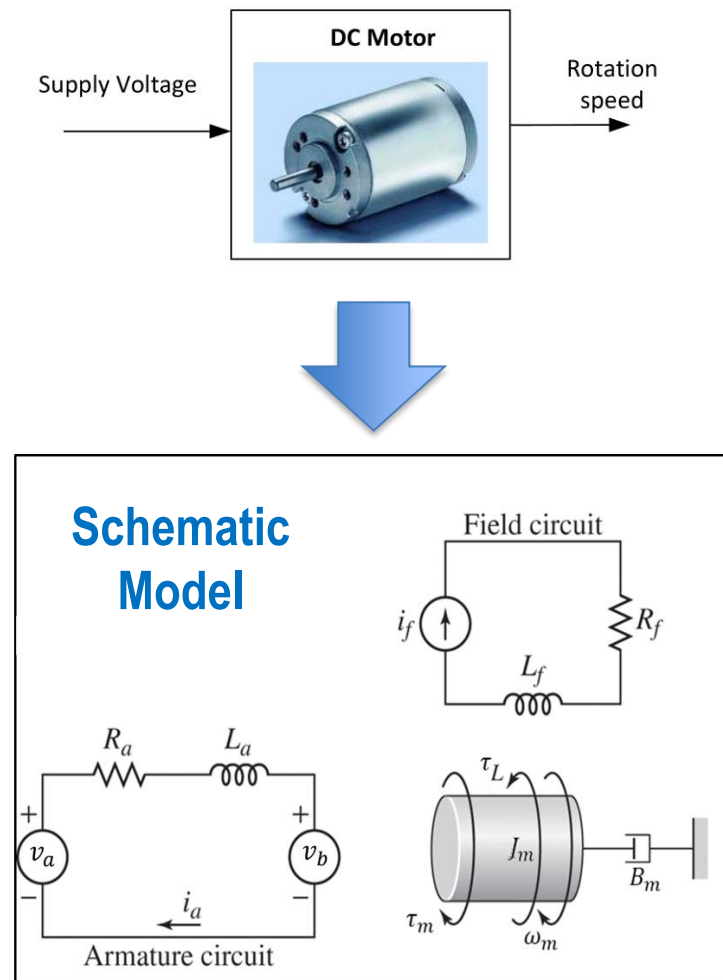


Simulink Model



System Simulation Tools: **Simscape**

- **Simcpace** is a MATLAB-based, object-oriented physical modeling language that enables the user to create models of physical components using an acausal modeling approach.
- This language is designed to use in MATLAB and Simulink environment; it can benefit MATLAB functions and Simulink blocks.



THANK YOU